

Assignments 1 & 2, 2026

Level: BE

Subject : Applied Mathematics

Due Date : May 31, 2026

Attempt all questions.

1. Show that the function $f(z) = z^2$ is entire function.
2. Check for analyticity of $f(z) = |z|^2 = z\bar{z}$.
3. Check the analyticity of $f(z) = \cos z$. Also, find $f'(z)$ if exists.
(Hints: if $f'(z)$ exists then $f'(z) = u_x + iv_x$)
4. Check the analyticity of $f(z) = xy + iy$,
5. Check the analyticity of $f(z) = (z^2 - 2)e^{-x}e^{-iy}$
6. check the analyticity of $f(z) = \bar{z}$
7. Check the analyticity of $f(z) = z^n$, where $n \in \mathbb{N}$. Also, find $f'(z)$ if exists.
(Hints: if $f'(z)$ exists then $f'(z) = f'(re^{i\theta}) = e^{-i\theta}(u_r + iv_r)$)
8. check the analyticity of $f(z) = z^{-3}$. Also, find $f'(z)$ if exists.
9. Check the analyticity of $f(z) = \sinh z$
10. Is $u(x, y) = 3x^2y + x^2 - y^3 - y^2$ harmonic function? If yes, find its harmonic conjugate v of u and also find corresponding analytic function $f(z) = u + iv$.
11. check $u = \sin x \cosh y$ is harmonic or not? If yes, find corresponding analytic function $f(z) = u + iv$.
12. Is $u = \arg z$ is harmonic? If yes, find a corresponding harmonic conjugate.
13. show that $v = 2xy - \frac{y}{x^2 + y^2}$ is a harmonic function. Also find the function u such that v is harmonic conjugate of u .
14. Is $u = \log|z|$ is a harmonic function? If yes, find its harmonic conjugate v and corresponding analytic function.
15. Given that $f(z) = u + iv$ is an analytic function and $u + v = e^x(\cos y + \sin y)$. Find $f(z)$.
16. Determine value of a , when $u = \cos ax \cosh 2y$, is harmonic and find its harmonic conjugate.
17. Show that bilinear transformation maps circles into circles.
18. Find the bilinear transformation which maps the points $z = i$, $z = -i$, $z = 1$ into the points $w = 0$, $w = 1$, $w = \infty$.
19. Show that the image of the circle $|z| = 1$ under the mobius mapping $w = \frac{5-4z}{4z-2}$ is $u^2 + v^2 + u = \frac{3}{4}$. Also, represent geometrically.
20. Integrate $\int_c ze^{z^2} dz$, where c from 1 along the axes to i .

21. Integrate $\int_c \bar{z} dz$, where c is a curve from 0 along the parabola $y = x^2$ to $1+i$.
22. Does Cauchy integral Theorem can be applicable to evaluate the integral $\int_c \frac{dz}{z^4 + \pi}$, where c is a unit circle $|z| = 1$? If possible, find the value.
23. Does Cauchy integral Theorem can be applicable to evaluate the integral $\int_c ze^{-z} dz$, where c is a unit circle $|z| = 1$? If possible, find the value.
24. Does Cauchy integral Theorem can be applicable to evaluate the integral $\int_c \log(1 - z) dz$, where c is the boundary of the parallelogram with vertices $i, -i, 1+i, -1-i$? If possible, find the value.
25. Does Cauchy integral Theorem can be applicable to evaluate the integral $\int_c \coth \frac{z}{2} dz$, where c is a circle $|z - \frac{\pi i}{2}| = 1$? If possible, find the value.
26. Does Cauchy integral Theorem can be applicable to evaluate the integral $\int_c (\bar{z})^3 dz$, where c is a unit circle $|z| = 1$?
27. Does Cauchy integral formula can be applicable to evaluate the integral $\int_c \frac{z+1}{z^3 - 4z} dz$, where c is a circle $|z - 2| = 3/2$? If possible, find the value.
28. Integrate $\int_c \frac{\cosh 3z}{5z} dz$, where c is a circle $|z| = 1$ (counterclockwise).
29. Integrate $\int_c \frac{\cos \frac{\pi z}{2n}}{z^{2n}} dz$, where c is a circle $|z| = 1$ (counterclockwise).
30. Integrate $\int_c \frac{z^2}{z^4 - 1} dz$, where c is a circle $|z + i| = 1$ (counterclockwise) by using Cauchy integral formula .
31. Integrate $\int_c \frac{\log(z-1)}{z-6} dz$, where c is a circle $|z - 6| = 4$ (counterclockwise).
32. Integrate $\int_c \frac{\sin z}{4z^2 - 8iz} dz$, where c consists of the boundaries of the square with vertices $3, -3, 3i, -3i$ (counterclockwise) and $1, -1, i, -i$ (clockwise).
33. Evaluate $\int_c \frac{z^6}{(2z-1)^6} dz$, where c is a circle $|z| = 1$ (counterclockwise).
34. Evaluate $\int_c \frac{\cot z}{(z-\pi/2)^2} dz$, where c is the ellipse $4x^2 + 9y^2 = 36$ (counterclockwise).
35. Evaluate $\int_c \frac{e^z}{(z-1)^2(z^2+4)} dz$, where c is a circle $|z| = 1.5$ (counterclockwise).
36. Evaluate $\int_c \frac{\cos z}{(z-\pi i)^2} dz$, where c any contour enclosing the point π (counterclockwise).
37. Evaluate $\int_c \frac{z+1}{z^3 - 2z^2} dz$, where c is a circle $|z| = 1$.
38. Integrate $\int_c \frac{z^3 + \sin z}{(z-i)^3} dz$, where c is the the boundary of the square with vertices $2, -2, 2i, -2i$ (counterclockwise).
39. Integrate $\int_c \frac{e^{z^2}}{z(z-2i)^2} dz$, where c consists of $|z - i| = 3$ (counterclockwise) and $|z| = 1$ (clockwise).
40. Find the Maculaurin's series of
 - (a) $f(z) = \sin^2 z$. Also, find the radius of convergence.
 - (b) $f(z) = \frac{z+2}{1-z^2}$. Also, find the radius of convergence.
 - (c) $f(z) = \frac{2-z}{(1-z)^2}$.
41. Find the Taylor's series of

- (a) $f(z) = e^z$ at $z = a$. Also, find the radius of convergence.
- (b) $f(z) = \sin z$ at $z = \pi/2$.
- (c) $f(z) = \frac{1}{z(z-1)}$ at $z = -1$. Also, find the circle and radius of convergence.
- (d) $f(z) = \frac{z^2-1}{(z+2)(z+3)}$ at $z = 0$. Also, find the circle and radius of convergence
42. Find the Taylor's or Laurent's series represent of the function $f(z) = \frac{1}{(1+z^2)(z+2)}$ when (i) $|z| < 1$
(ii) $1 < |z| < 2$ (iii) $|z| > 2$.
43. Find the Taylor's or Laurent's series represent of the function $f(z) = \frac{z+3}{z(z^2-z-2)}$ when (i) $0 < |z| < 1$
(ii) $1 < |z| < 2$ (iii) $|z| > 2$.
44. Find the Taylor's or Laurent's series represent of the function $f(z) = \frac{1}{z-z^3}$ when (i) $1 < |z-1| < 1$
(ii) $1 < |z+1| < 2$ (iii) $|z+1| > 2$.
45. Find the Taylor's or Laurent's series represent of the function $f(z) = \frac{1}{z+z^3}$ when (i) $1 < |z-i| < 2$
(ii) $1 < |z+i| < 2$ (iii) $|z+i| > 2$.
46. Find the Taylor's or Laurent's series represent of the function $f(z) = \frac{2z-3i}{z^2-3iz-2}$ when (i) $|z| < 1$ (ii) $1 < |z| < 2$ (iii) $|z| > 2$.
[Hint: $z^2 - 3iz - 2 = z^2 - 2iz - iz + i^2 2 = (z-2i)(z-i)$]

"ALL THE BEST."